Available software

The software should be able to tell us what containers are running, provide a live view of the resources consumed within the containers, and give statistical analysis.

The first set of software that addressed this opportunity were the server monitoring frameworks that added on a Docker monitoring plugin. The Scout server monitoring framework repurposed its LXC monitoring solution to Docker containers. This was the approach adopted by open source equivalents like Zabbix and Icinga as well.

Commercial offerings that showed up included DataDog, a monitoring service that added Docker monitoring capability to its myriad enterprise monitoring and visualisation features.

Agent-driven open source general-purpose monitoring frameworks like 'host sFlow' released support for Docker containers, exporting standard sFlow performance metrics for LXC containers and unifying it as part of the sFlow ecosystem that encompasses applications, virtual servers, virtual networks, servers and networks.

cAdvisor

cAdvisor or container advisor is open source software that has native support for Docker containers, and is designed as a daemon that collects, aggregates, processes and exports information about running containers.

In this write-up, we won't go into the details of how to compile or build cAdvisor; instead, we will focus on using a ready-to-run cAdvisor image from *hub.docker.com* and then run the image.

First, we need to set up the Docker environment and then enable it using *systemctl*. If you are on a distro that does not use *systemd*, then */etc/init.d/docker* would be the most common command to start Docker.

```
$ sudo systemctl status docker
root's password:
docker.service - Docker
  Loaded: loaded (/usr/lib/systemd/system/docker.service;
disabled)
   Active: inactive (dead)
$ sudo systemctl enable docker
$ sudo systemctl status docker
docker.service - Docker
   Loaded: loaded (/usr/lib/systemd/system/docker.service;
enabled)
   Active: inactive (dead)

$ sudo systemctl start docker
$ sudo systemctl status docker
docker.service - Docker
   Loaded: loaded (/usr/lib/systemd/system/docker.service;
enabled)
   Active: active (running) since Wed 2015-04-08 21:15:59
IST; 14h ago
 Main PID: 14880 (docker)
   CGroup: /system.slice/docker.service
           └─14880 /usr/bin/docker -d

Apr 08 21:15:59 beijing.site docker[14880]: time="2015-04-
08T21:15:59+05:30" level="info" msg="+job serveapi(unix:///
var/run/...sock)"
Apr 08 21:15:59 beijing.site docker[14880]: time="2015-
04-08T21:15:59+05:30" level="info" msg="+job init_
networkdriver()"
Apr 08 21:15:59 beijing.site docker[14880]: time="2015-04-
08T21:15:59+05:30" level="info" msg="Listening for HTTP on
unix (/v...sock)"
Apr 08 21:16:00 beijing.site docker[14880]: time="2015-
04-08T21:16:00+05:30" level="info" msg="-job init_
networkdriver() = OK (0)"
Apr 08 21:16:00 beijing.site docker[14880]: time="2015-04-
08T21:16:00+05:30" level="info" msg="WARNING: Your kernel
does not ...imit."
Apr 08 21:16:00 beijing.site docker[14880]: time="2015-04-
08T21:16:00+05:30" level="info" msg="Loading containers:
start."
Apr 08 21:16:00 beijing.site docker[14880]: time="2015-04-
08T21:16:00+05:30" level="info" msg="Loading containers:
done."
Apr 08 21:16:00 beijing.site docker[14880]: time="2015-04-
08T21:16:00+05:30" level="info" msg="docker daemon: 1.5.0
a8a31ef; ...btrfs"
Apr 08 21:16:00 beijing.site docker[14880]: time="2015-
04-08T21:16:00+05:30" level="info" msg="+job
acceptconnections()"
Apr 08 21:16:00 beijing.site docker[14880]: time="2015-04-
08T21:16:00+05:30" level="info" msg="-job acceptconnections()
= OK (0)"
Apr 08 21:17:28 beijing.site docker[14880]: time="2015-04-
08T21:17:28+05:30" level="info" msg="GET /v1.17/version"
Apr 08 21:17:28 beijing.site docker[14880]: time="2015-04-
08T21:17:28+05:30" level="info" msg="+job version()"
Apr 08 21:17:28 beijing.site docker[14880]: time="2015-04-
08T21:17:28+05:30" level="info" msg="-job version() = OK (0)"
Hint: Some lines were ellipsized, use -l to show in full.
```

Once you have verified that the Docker service is running fine, you need to add yourself to the Docker group.

```
$ sudo usermod -a -G docker saifi
```

To verify which groups you are part of, just issue the following command:

```
$ groups
```